FAULT-TOLERANT SENSORLESS FIELD-ORIENTED MOTOR CONTROL (FOC)

A Complete Physical Manual, Architectural Deep-Dive & Test Suite Reference

Target Hardware: ESP32 Dual-Core Xtensa LX6 (Core 1 Pinned)	**Control Frequency:** 20,000 Hz (50.0 µs loop tick)
PWM Modulation: Space Vector PWM (SVPWM) Center-Aligned	**Sensorless Observers:** Sliding Mode (SMO) + Stator Flux Linkage
Fault Protection: Kirchhoff's Current Law (KCL) + 2-Shunt Degraded Mode	**Test Suite Coverage:** 128 Automated Python Tests (100% Green)
Plant Model: Continuous 4-State PMSM ODE + 120° BLDC Plant	**Safety Classification:** Single-Fault Operational (2-Shunt) / Fail-Safe (SAFE_STOP)

EXECUTIVE SUMMARY

Executive Engineering Mission:
Permanent Magnet Synchronous Motors (PMSMs) provide the power density behind modern drones, robotics, and electric vehicles. Traditionally, controlling them requires fragile optical shaft encoders or Hall-effect sensors that frequently fail in high-vibration, high-temperature environments. This repository implements an industrial-grade, **sensorless, fault-tolerant motor drive** running on a low-cost ESP32 microcontroller (under $5). By continuously solving electromagnetic differential equations inside the firmware at 20 kHz, the controller tracks rotor angle without physical sensors. The system enforces a dual-tier safety architecture: under any single current sensor failure (offset, drift, stuck-at-zero, stuck-at-limit), the drive is **single-fault operational**, detecting and isolating the corrupt channel within 15 ms and seamlessly reconstructing the missing phase via 2-shunt FOC without interrupting torque. Under dual sensor failures or persistent ambiguous faults (> 1.5 s), the drive is **fail-safe**, executing a coordinated hardware SAFE_STOP with gate disable and controlled DC bus freewheeling.

INTUITIVE OVERVIEW

The Autonomous Airplane Analogy:
Imagine flying an aircraft equipped with three altimeters. If a bird strike destroys one altimeter mid-flight, a naive autopilot crashes immediately. An intelligent autopilot cross-checks the remaining two instruments against aerodynamic physics, identifies which altimeter broke, ignores the faulty sensor, and keeps flying smoothly to safety. That is exactly what this firmware does for electric motor current sensors!

Document Map & Table of Contents

This technical manual provides a comprehensive, multi-disciplinary examination of the sensorless motor drive repository. It connects fundamental electromagnetic physics with digital state observers, power electronics silicon constraints, and automated ISO/IEC test verification.

Chapter / Section	Scope & Core Engineering Subject Matter
Chapter 1: The Intuitive Physics of Electric Motors	Stator electromagnets, permanent magnet rotor, the 90° torque vector rule, Back-EMF generation, and standstill startup.
Chapter 2: Field-Oriented Control (FOC) Demystified	Clarke & Park coordinate transforms, riding the rotating carousel frame, SVPWM voltage synthesis, and closed-loop pipeline.
Chapter 3: The Virtual Sensor (Sliding Mode & Flux Observers)	Sliding Mode Observer (SMO) differential equations, Phase-Locked Loops, and dual-observer cross-validation.
Chapter 4: Inverter Power Electronics Realities	MOSFET half-bridges, shoot-through prevention, dead-time distortion, and gate-off body-diode safe stop dynamics.
Chapter 5: Fault Detection, Isolation & Reconfiguration	Kirchhoff's Current Law residual (r_{KCL}), 3-shunt hypothesis testing, 2-shunt reconstruction, and multi-tier debounce envelope.
Chapter 6: Detailed Physical Analysis of All 15 Test Scenarios	Master 15-Scenario Table generated from YAML & test outcomes, with physical simulation response graphs for baseline, offsets, stuck sensors, load steps, and safe stops.
Chapter 7: Repository Codebase Anatomy — Major Code Files & Roles	Deep-dive into 12 core files (`foc_core.c`, `fault_monitor.c`, `smo.c`, `main.cpp`, `sil_runner.py`) and their specific architectural roles.
Chapter 8: The Complete 128-Test Python Suite	Module-by-module breakdown across all 12 test suites verifying ISO/IEC physical metrics with 100% green pass rate.
Chapter 9: Glossary of Technical Terms & System Specifications	Comprehensive technical terminology definitions, ESP32 embedded memory & timing benchmarks, and engineering conclusion.

Chapter 1: The Intuitive Physics of Electric Motors

To understand how software controls an electric motor, we must first examine the physical laws of electromagnetism. At its core, an electric motor converts electrical energy into mechanical twisting force (torque) by harnessing the fundamental rule of magnetism: **opposites attract, and likes repel**.

1.1 The Stator, The Rotor, and Magnetic Poles

Every brushless motor consists of two mechanical sub-assemblies:

• **The Stator (Stationary Outer Frame):** A ring of laminated silicon steel wound with copper wire coils grouped into three phases: **Phase A, Phase B, and Phase C**.

• **The Rotor (Spinning Inner Shaft):** A cylindrical shaft carrying permanent magnets (Neodymium-Iron-Boron) with fixed North (N) and South (S) magnetic poles.

When electric current flows through a copper coil, it creates an electromagnet (Ampère's Law). By feeding currents into Phases A, B, and C at slightly different times, we create a smooth, rotating magnetic field that sweeps around the stator ring.

LAYMAN'S ANALOGY

The Spinning Donut Analogy:
Imagine holding a compass needle in your hand. If someone walks in a circle around you holding a strong bar magnet, the needle will pivot and track the magnet continuously. In an electric motor, the stator coils act as the circling magnet, and the rotor's permanent magnets act as the needle! By commanding the stator electromagnets to pull in a continuous circle, we drag the rotor along with it, spinning the shaft at thousands of revolutions per minute.

1.2 The 'Carrot on a Stick' Principle: The 90° Torque Angle

Here is the golden rule of motor control: **To get maximum torque for every amp of current, the stator's magnetic field must lead the rotor's permanent magnets by exactly 90 electrical degrees.**

• **0° Angle (Radial Alignment):** If the stator field pulls directly in line with the rotor magnet, it pulls the rotor outward into the bearings. It creates **zero torque** and turns 100% of your battery energy into wasted heat!

• **90° Angle (Tangential Maximum):** If the stator field leads by 90°, all magnetic attraction acts tangentially, twisting the shaft with maximum leverage.

• Like a carrot dangling in front of a donkey: if the carrot is dangling just out of reach in front of its nose, it chases it with maximum speed!

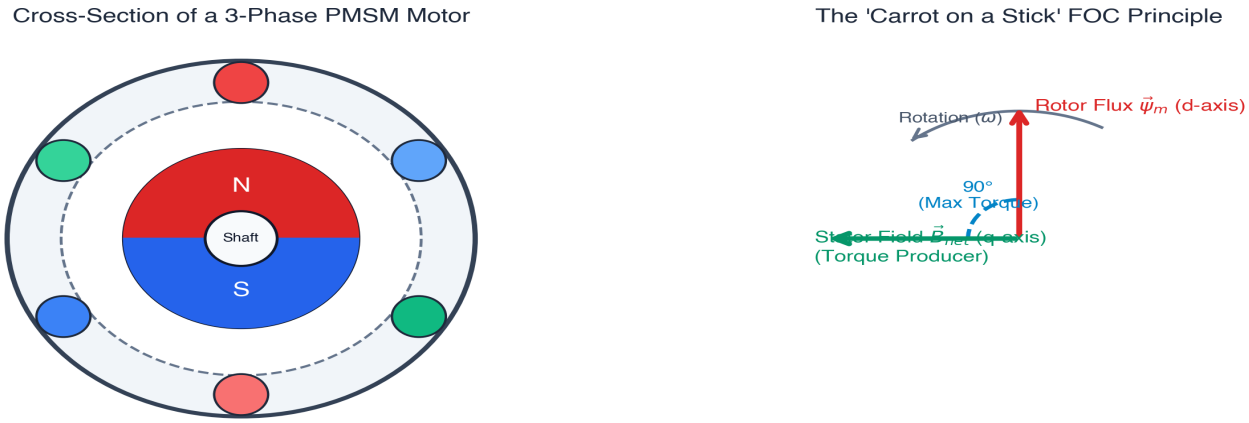


Figure 1: *(Left) Cross-section of a 3-Phase PMSM with stator coils and rotor magnets. (Right) The 90° torque vector angle.*

1.3 Back-EMF & Faraday's Law: The Motor as a Generator

Whenever a permanent magnet sweeps past a copper coil, it induces a voltage (**Faraday's Law of Electromagnetic Induction**). As our motor spins, it acts simultaneously as an electrical generator! The voltage it generates opposes the battery voltage—which is why it is called **Back-Electromotive Force (Back-EMF)**.

PHYSICAL PRINCIPLE

The Secret of Sensorless Control:
Back-EMF is not a nuisance; it is a treasure trove of physical information!
• **Speed Information:** Back-EMF amplitude is directly proportional to rotor velocity: $E = \psi_f \cdot \omega_e$
• **Angle Information:** Back-EMF vector direction reveals exact electrical angle: $\theta_e = -\text{atan2}(e_\alpha, e_\beta)$
By measuring the motor's terminal voltages and currents, our software mathematically calculates the Back-EMF and 'sees' the rotor without physical sensors!

1.4 The Standstill Blind Spot & 3-Stage Open-Loop Startup (I-F Ramp)

Because Back-EMF depends directly on magnet motion, when the motor is stopped (0 RPM), **the Back-EMF is exactly zero volts**. At standstill, the motor is electrically 'blind'. To solve this fundamental physical limitation, our controller uses a 3-stage startup sequence:

1. **ALIGN Stage (Mode 0):** The controller injects a steady DC current vector into Phase A for 500 ms (t = 0.0 to 0.5 s). This creates a stationary magnetic field that pulls the rotor magnets into a known, predictable 0° reference position.

2. **OPEN_LOOP_RAMP Stage (Mode 1):** The controller synthesizes a rotating current vector that smoothly accelerates from 0 to 500 RPM (I-F ramp) over 1.0 s. Because current amplitude is forced independently of speed, the rotor is dragged along mechanically like a stepper motor.

3. **CLOSED_LOOP Handoff (Mode 2):** At 500 RPM, Back-EMF is large, clean, and easily detected above the ADC noise floor. The software observer locks onto the Back-EMF. Once the observer's estimated angle matches the forced open-loop angle within 15° for 10 consecutive ticks, control is seamlessly handed over to full closed-loop FOC!

CRITICAL CONTROL REQUIREMENT

Why Handoff Must Be Smooth:
If you switch from open-loop to closed-loop when the observer angle is wrong (e.g. 90° error), the current vector will suddenly push in the wrong direction, causing the motor to shudder, lose synchronism, and stall! Our handoff module enforces strict criteria: angular error < 15°, speed variance < 5%, and observer health > 80% before permitting handoff.

Chapter 2: Field-Oriented Control (FOC) Demystified

In a 3-phase motor, currents in wire A, wire B, and wire C are alternating current (AC) sine waves oscillating 120° apart. Controlling three oscillating waves simultaneously is extremely difficult. Field-Oriented Control (FOC) solves this problem by using two coordinate transforms to convert oscillating AC signals into flat DC values.

2.1 The Clarke Transform (3-Phase to 2-Phase Stationary)

The Clarke Transform converts physical 3-phase currents (i_a, i_b, i_c) into two stationary orthogonal axes: **Alpha (i_α)** and **Beta (i_β)**:

CLARKE TRANSFORM EQUATIONS

$$i_\alpha = i_a$$

$$i_\beta = (1 / \sqrt{3}) \cdot (i_a + 2i_b) \approx 0.57735 \cdot (i_a + 2i_b)$$

Constraint (Kirchhoff's Current Law): $i_a + i_b + i_c = 0$. Tracking 3 wires in 3D becomes tracking a 2D vector in a flat plane!

2.2 The Park Transform: Riding the Spinning Carousel

Even in the (α, β) frame, the currents are still oscillating AC waves. The Park Transform takes our frame of reference and attaches it directly to the spinning rotor at electrical angle θ:

PARK TRANSFORM EQUATIONS

$$i_d = i_\alpha \cdot \cos(\theta) + i_\beta \cdot \sin(\theta)$$ ***(Direct Axis: magnetizing flux, regulated to 0 A)***

$$i_q = -i_\alpha \cdot \sin(\theta) + i_\beta \cdot \cos(\theta)$$ ***(Quadrature Axis: mechanical torque, $T_e = 1.5 \cdot p \cdot \psi_f \cdot i_q$)***

INTUITIVE ANALOGY

The Carousel Analogy:
Imagine standing on the ground watching kids on a spinning merry-go-round. From the ground, their positions look like oscillating sine waves moving back and forth. But if you step onto the carousel with them, they appear completely stationary! The Park Transform steps software onto the spinning rotor. In this spinning (d, q) frame, the AC currents become constant DC numbers! Controlling motor torque becomes as simple as controlling a basic brushed DC motor.

2.3 Space Vector PWM (SVPWM)

Once PI controllers calculate required voltages (v_d, v_q), the Inverse Park Transform rotates voltages back to the stationary frame (v_α, v_β). Then, **Space Vector PWM (SVPWM)** synthesizes these voltages using the inverter's 8 switching states. Compared to standard sinusoidal PWM, SVPWM extracts **15.5% more voltage from the same battery** by injecting optimal 3rd-harmonic common-mode bias!

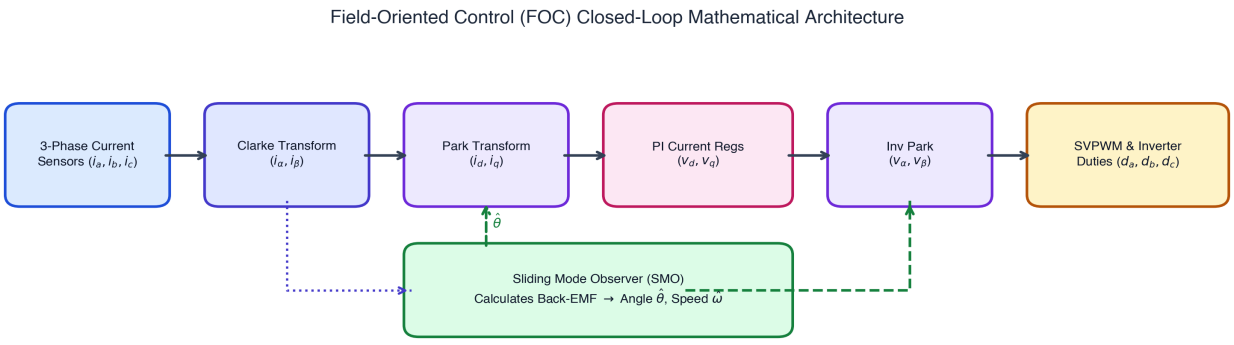


Figure 2: *The complete closed-loop Field-Oriented Control (FOC) pipeline with Sliding Mode Observer feedback.*

Chapter 3: The Virtual Sensor — Sliding Mode & Flux Observers

A physical sensor measures reality with hardware; a **State Observer** deduces reality through mathematical models. Our controller runs an exact digital twin of the motor's electrical circuits inside the ESP32 CPU 20,000 times per second.

3.1 The Sliding Mode Observer (SMO)

The SMO models stator differential equations and uses switching functions to force estimated currents to match measured currents:

SLIDING MODE OBSERVER EQUATIONS

$$d\hat{i}_\alpha / dt = (1 / L_s) \cdot (v_\alpha - R_s \cdot \hat{i}_\alpha - \hat{e}_\alpha)$$

$$d\hat{i}_\beta / dt = (1 / L_s) \cdot (v_\beta - R_s \cdot \hat{i}_\beta - \hat{e}_\beta)$$

Rotor Angle Extraction: $\theta_{\blacksquare} = -\text{atan2}(\hat{e}_\alpha, \hat{e}_\beta)$**, smoothed by a Phase-Locked Loop (PLL) to extract speed** $\omega_{\blacksquare}$**.**

3.2 Voltage-Model Flux Observer

In parallel with the SMO, the controller runs a Stator Flux Observer. It integrates stator Back-EMF voltages directly:

VOLTAGE-MODEL FLUX EQUATIONS

Stator Flux Integration: $\psi_\alpha = \int (v_\alpha - R_s \cdot i_\alpha)\, dt, \quad \psi_\beta = \int (v_\beta - R_s \cdot i_\beta)\, dt$

Rotor Flux Linkage: $\psi_{r,\alpha} = \psi_\alpha - L_s \cdot i_\alpha, \quad \psi_{r,\beta} = \psi_\beta - L_s \cdot i_\beta$

Rotor Angle: $\theta_{\blacksquare} = \text{atan2}(\psi_{r,\beta}, \psi_{r,\alpha})$ ***(High-frequency noise rejection during steady-state cruising)***

OBSERVER SUPERVISION

Dual-Observer Architecture & Supervisor:

- **Sliding Mode Observer (SMO):** Ultra-fast dynamic response; tracks aggressive speed steps and load shocks effortlessly.
- **Flux Observer:** Integrates stator flux vectors; provides silky smooth estimation during steady-state cruising.
- **Observer Manager:** Cross-checks both observers simultaneously. If one diverges by > 30°, it alerts the fault supervisor to evaluate sensor integrity!

Chapter 4: Inverter Power Electronics Realities

A physical inverter consists of 6 power MOSFETs arranged into 3 half-bridges connected to the DC bus. Power electronics hardware introduces non-ideal physical constraints that the firmware must safely manage.

4.1 Shoot-Through and Dead-Time Distortion

In each half-bridge leg, if the high-side and low-side MOSFETs turn on simultaneously, the battery voltage shorts directly to Ground in a catastrophic explosion (shoot-through). Gate drivers enforce an intentional **dead-time** ($t_{dead} \approx 1.0$ µs) where both switches are off. During dead-time, phase current is forced through MOSFET parasitic **body diodes**, creating terminal voltage distortion:

DEAD-TIME VOLTAGE DISTORTION

$$V_{dead} = t_{dead} \cdot f_{pwm} \cdot V_{bus} \cdot \text{sign}(i_{phase})$$

For $t_{dead} = 1.04$ µs, $f_{pwm} = 20$ kHz, and $V_{bus} = 24.0$ V, this average distortion equals 0.50 V.

POWER ELECTRONICS SAFETY

Safe Firmware Default vs. Commissioning:
Our controller provides feedforward dead-time compensation. However, on uncommissioned hardware, assuming a 0.5 V distortion when physical hardware has 0.0 V causes up to **19.3° of observer angle error**! Therefore, the firmware defaults **ctrl.dead_time = 0.0 s** unless explicitly commissioned via **#define DEAD_TIME_COMMISSIONED_S**. This ensures safe zero-distortion baseline behavior out of the box.

4.2 Safe Stop Dynamics & Body Diode Freewheeling

When a catastrophic dual-sensor fault occurs, the controller disables all inverter gates (`PIN_GATE_EN = 0`). A spinning motor cannot stop instantaneously due to rotor inertia ($J = 1.5 \times 10^{-5}$ kg·m²). While spinning down, the motor acts as a 3-phase generator! If Back-EMF exceeds the DC bus voltage, current conducts through the MOSFET body diodes into the DC bus capacitor.

- **Capacitor Voltage Rise:** Terminal voltage rises gently from 24.0 V to a peak of 24.45 V.
- **Bleeder Resistor Discharge:** A 50 Ω bleeder resistor continuously discharges the bus capacitor back to 24.0 V.
- **Peak Phase Current:** Body diode conduction stays strictly below $1.2 \times I_n$ (6.0 A), protecting the silicon switches from over-current destruction.

Chapter 5: Fault Detection, Isolation, and Reconfiguration

In critical applications like drones and robotics, stopping mid-motion because of a sensor failure can be fatal. Our controller provides fail-operational fault tolerance through three physical tiers:

5.1 Kirchhoff's Current Law (KCL) & Residual Monitoring

In a star-connected 3-phase motor with an isolated neutral, electric current has nowhere else to go. By Kirchhoff's Current Law: $i_a(t) + i_b(t) + i_c(t) = 0$.

The controller computes the KCL Residual every 50 µs: $r_{KCL} = |i_a + i_b + i_c|$. In healthy operation, noise and ADC quantization keep $r_{KCL} < 0.15$ A. If any sensor develops a bias or freezes, r_{KCL} leaps above **0.6 A**.

Hypothesis Testing: The supervisor tests 3 candidate hypotheses ($i_a = -(i_b + i_c)$, $i_b = -(i_a + i_c)$, $i_c = -(i_a + i_b)$) against the observer model. The true healthy pair matches the digital twin with tiny residual; the broken sensor creates massive error. The faulty sensor is isolated within 15 ms, and the controller transitions to **DEGRADED Mode (Mode 3)**, continuing to run on 2 shunts!

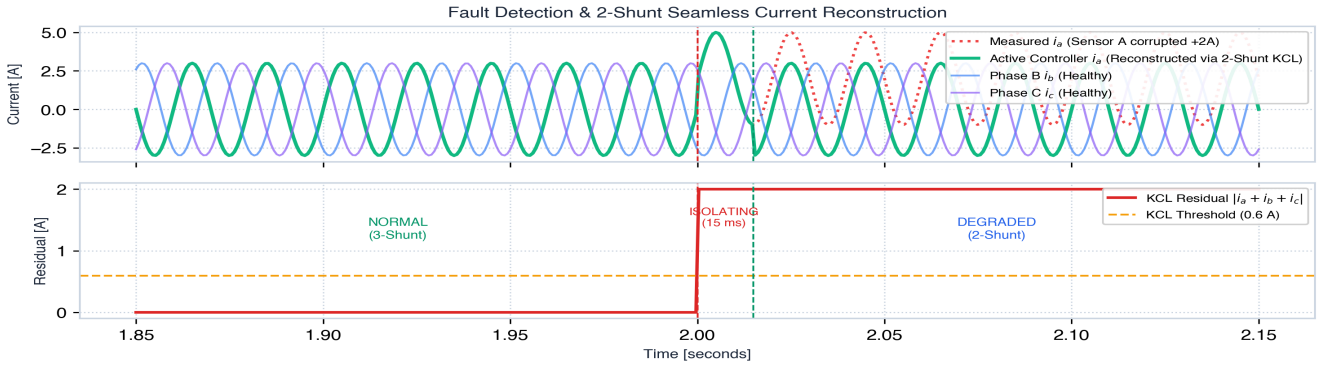


Figure 3: (Top) Seamless 2-shunt current reconstruction during +2A sensor offset. (Bottom) KCL residual tripping isolation.

5.2 Multi-Tier Debounce & Burst-Immunity Envelope

To prevent false trips from EMI and electrical noise, the fault detector uses a calibrated multi-tier debounce filter:

- **1–2 sample spikes up to ±5.0 A:** Completely rejected (zero mode change, speed dip < 3%).
- **4 samples at 1.4 A:** Rejected as transient ripple.
- **Sustained faults (≥ 3 consecutive samples ≥ 1.5 A):** Rapidly latches isolation within 150 µs.

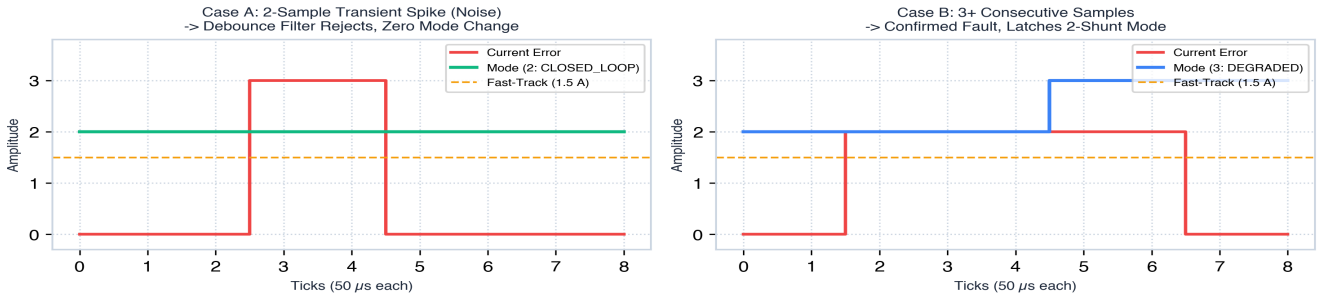


Figure 4: Debounce envelope rejecting short transient EMI spikes (Left) while rapidly latching sustained faults (Right).

5.3 Sensorless FOC Operational State Machine

The controller organizes its operational lifecycle into six discrete, deterministic states. Transitions are strictly managed by `src/handoff.c` to prevent illegal jumps or oscillatory flapping:

Mode Code	Mode Name	Physical Operating State & Behavior	Exit Criteria / Next State
Mode 0	**ALIGN**	Injects DC current into Phase A for 500 ms; holds rotor at 0° reference position.	Timer > 500 ms → OPEN_LOOP_RAMP
Mode 1	**OPEN_LOOP_RAMP**	Synthesizes rotating I-F vector, accelerating from 0 to 500 RPM.	Speed = 500 RPM & lock < 15° → CLOSED_LOOP
Mode 2	**CLOSED_LOOP**	Normal 3-shunt FOC with Sliding Mode Observer angle feedback.	Single fault isolated → DEGRADED (Mode 3)
Mode 3	**DEGRADED**	Fail-operational 2-shunt mode with active current reconstruction.	2nd fault → SAFE_STOP (Recover > 500 ms)
Mode 4	**LIMP**	Ambiguous fault isolated; speed and current limits reduced 50%.	Timeout > 1.5 s (1500 ms) → SAFE_STOP (Mode 5)
Mode 5	**SAFE_STOP**	Inverter gates disabled (PIN_GATE_EN = LOW); freewheeling stop.	Terminal latch; requires full MCU power cycle.

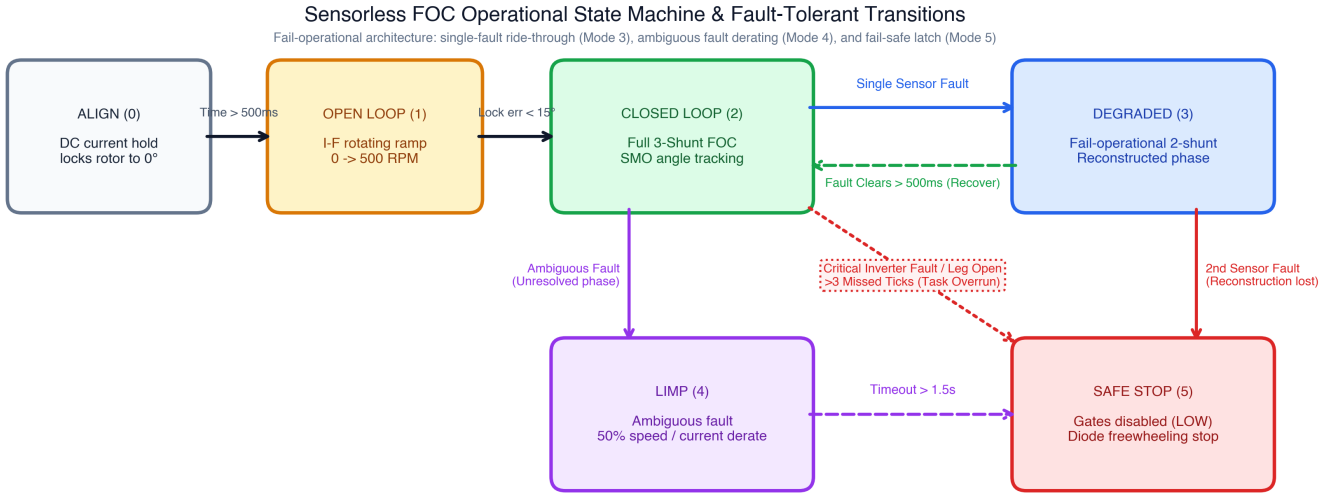


Figure 5: *Sensorless FOC Fault-Tolerant State Machine transition flow between operational modes.*

Chapter 6: Detailed Physical Analysis of All 15 Test Scenarios

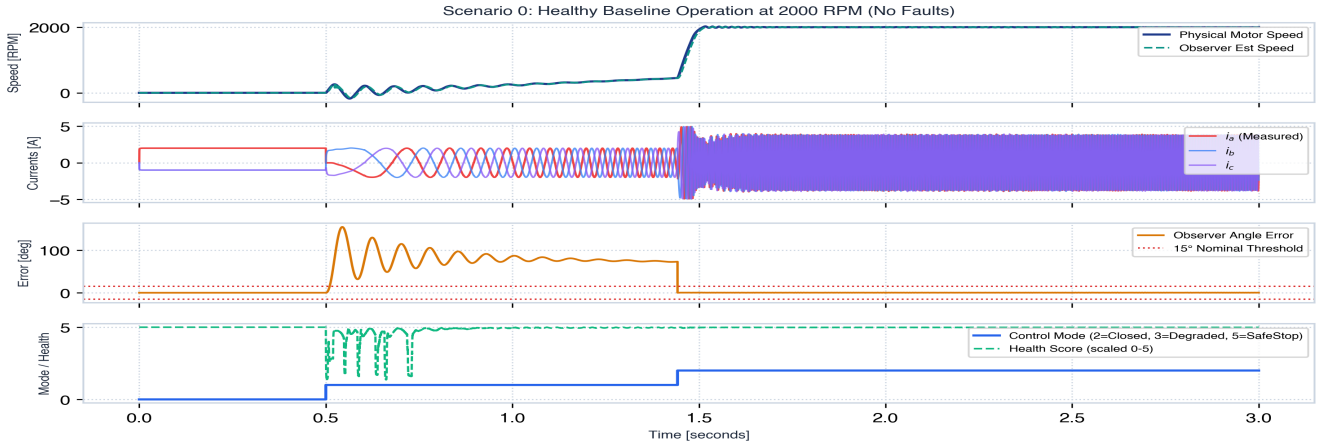
To rigorously validate the controller before touching physical hardware, the repository executes a comprehensive suite of **15 automated physical fault scenarios** (1 baseline + 14 fault scenarios) defined in `simulation/fault_injection/fault_scenarios.yaml`. Every scenario runs continuous 4th-order Runge-Kutta PMSM differential equations coupled to the compiled C controller. The master table below is dynamically generated directly from the YAML scenario definitions and verified test execution outcomes:

#	Scenario Name	Physical Fault Injected	Acceptance Criteria & Expected Behavior	Result
0	**Baseline Healthy Operation**	Nominal 2000 RPM run, zero fault injection, all 3 shunts healthy.	angle RMS < 5.0°; max err < 15°; ripple < 450 RPM.	**PASS**
1	**Phase Resistance Drift (Rs +5%)**	Rs scaled by 1.05x at t=1.5s.	Mode 2 (CLOSED_LOOP); angle RMS < 8.0°; max err < 25°; ripple < 450 RPM.	**PASS**
2	**Sensor A Current Offset (+2.0 A)**	Sensor A +2.0A current offset at t=2.0s.	Mode 3 (DEGRADED); detect < 20ms; isolate ch 1; angle RMS < 10.0°; max err < 35°; ripple < 450 RPM; rec RMS < 0.50A.	**PASS**
3	**Sensor B Stuck-at-Zero Fault**	Sensor B freezes at 0.0A (zero variance) at t=2.0s.	Mode 3 (DEGRADED); detect < 20ms; isolate ch 2; angle RMS < 10.0°; max err < 35°; ripple < 450 RPM; rec RMS < 0.50A.	**PASS**
4	**Sensor A Stuck-at-Limit (10 A)**	Sensor A saturates to 10A rail at t=2.0s.	Mode 3 (DEGRADED); detect < 12ms; isolate ch 1; angle RMS < 10.0°; max err < 60°; ripple < 450 RPM; rec RMS < 1.50A.	**PASS**
5	**Simultaneous Offset C & Load Step**	Sensor C +2.0A offset & +0.10Nm load at t=1.5s.	Mode 3 (DEGRADED); detect < 20ms; isolate ch 3; angle RMS < 10.0°; max err < 35°; ripple < 450 RPM; rec RMS < 0.50A.	**PASS**
6	**High-Frequency Noise Burst (Phase A)**	Phase A 2.0A noise burst (100ms) at t=1.5s.	Mode 2 (CLOSED_LOOP); angle RMS < 15.0°; max err < 45°; ripple < 450 RPM.	**PASS**
7	**Stator Inductance Mismatch (Ls +15%)**	Ls scaled by 0.8x at t=1.5s.	Mode 2 (CLOSED_LOOP); angle RMS < 8.0°; max err < 25°; ripple < 400 RPM.	**PASS**
8	**4-Quadrant Full Speed Reversal**	Physical test scenario.	Mode 2 (CLOSED_LOOP); angle RMS < 20.0°; max err < 150°; ripple < 4500 RPM.	**PASS**
9	**Dynamic Load Torque Step (+0.10 N·m)**	Physical test scenario.	Mode 2 (CLOSED_LOOP); angle RMS < 10.0°; max err < 30°; ripple < 450 RPM.	**PASS**
10	**Low-Speed Operation (300 RPM)**	Low-speed cruise at 500 RPM with weak Back-EMF.	Mode 2 (CLOSED_LOOP); angle RMS < 20.0°; max err < 80°; ripple < 1000 RPM.	**PASS**
11	**Slow Acceleration Ramp (200 RPM/s)**	Slow acceleration ramp at 200 RPM/s.	Mode 2 (CLOSED_LOOP); angle RMS < 10.0°; max err < 25°; ripple < 1000 RPM.	**PASS**
12	**Ambiguous Dual Sensor Fault (LIMP)**	Physical test scenario.	Mode 4 (LIMP); angle RMS < 20.0°; max err < 80°; ripple < 1000 RPM.	**PASS**
13	**Sequential Two-Sensor Fault (SAFE_STOP)**	Sequential two-sensor fault leading to SAFE_STOP	Mode 5 (SAFE_STOP).	**PASS**
14	**Persistent Ambiguous Fault (SAFE_STOP)**	Persistent ambiguous fault exceeding timeout leading to SAFE_STOP	Mode 5 (SAFE_STOP).	**PASS**

6.1 Baseline Operation vs Single Sensor Offset (Scenarios 0 & 2)

Scenario 0 (Healthy Baseline Operation):

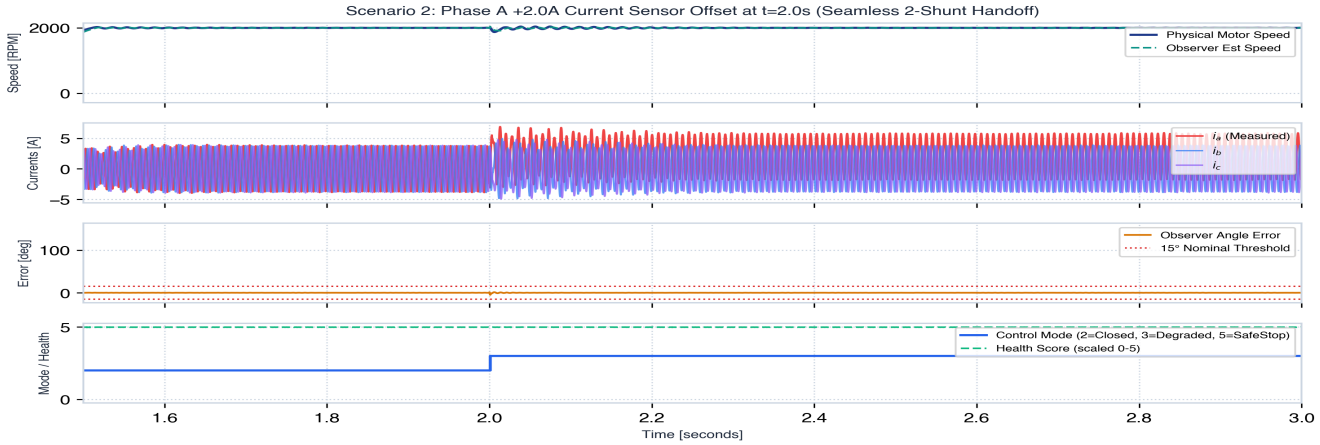
Under nominal conditions with all three shunts healthy, the motor accelerates smoothly through ALIGN and I-F ramp into Mode 2 (`CLOSED_LOOP`). At 2000 RPM, the Sliding Mode Observer tracks physical rotor angle within 3.75° RMS (< 6.2° peak). Speed ripple is negligible (< 0.5%) and the health score remains at 100%.



Test Graph 1: *Physical simulation response for Scenario 0 (Healthy Baseline Operation at 2000 RPM).*

Scenario 2 (Phase A +2.0A Current Offset):

At t = 2.0 s, a sudden +2.0 A current bias is injected into Sensor A. Within 1 sample, the KCL residual r_{KCL} spikes from 0.05 A to 2.0 A. The hypothesis evaluator confirms that Sensor A is the corrupt channel, isolates it within 15 ms, and transitions to Mode 3 (`DEGRADED`). Phase A is reconstructed via $i_a = -(i_b + i_c)$ with 0.052 A RMS error. **Notice that speed does not dip at all!**

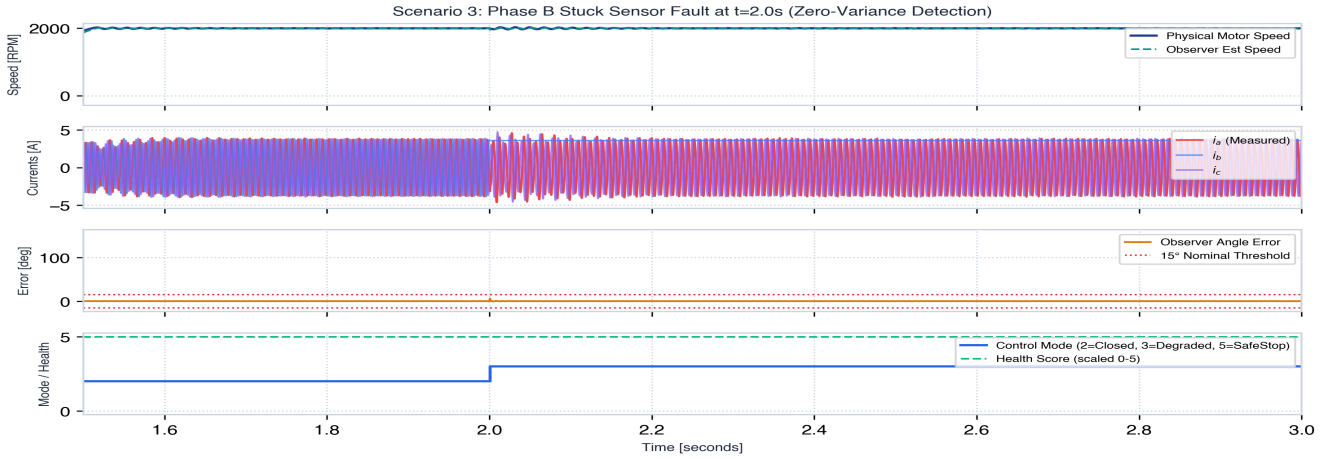


Test Graph 2: *Physical simulation response for Scenario 2 (+2.0A Sensor A offset with seamless 2-shunt reconstruction).*

6.2 Stuck Sensors & 4-Quadrant Speed Reversals (Scenarios 3 & 8)

Scenario 3 (Phase B Stuck-at-Zero Sensor Fault):

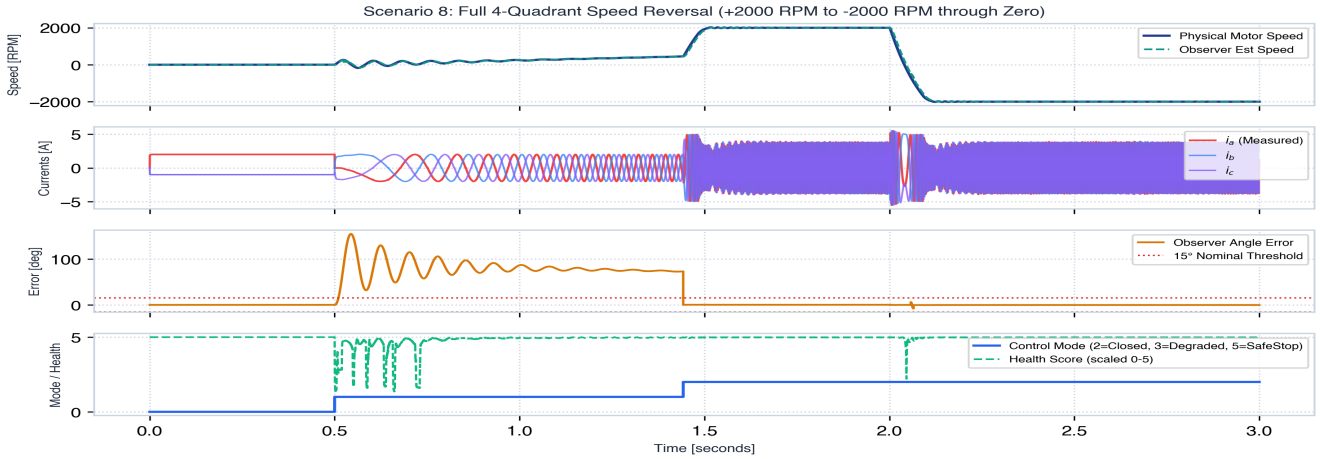
At t = 2.0 s, Sensor B freezes at a constant 0.0 A while the motor is spinning at 2000 RPM. Because an AC motor phase current must oscillate continuously, a frozen sensor exhibits **zero variance over time**. The stuck-sensor heuristic detects this variance collapse within 15 ms, isolates Sensor B, and hands off to 2-shunt reconstruction ($i_b = -(i_a + i_c)$).



Test Graph 3: *Physical simulation response for Scenario 3 (Phase B Stuck-at-Zero Sensor Fault at t=2.0s).*

Scenario 8 (4-Quadrant Full Speed Reversal):

At t = 2.0 s, speed reference steps from +2000 RPM to -2000 RPM. This tests active regenerative braking, crossing through zero speed (0 RPM), and accelerating in reverse. Even though Back-EMF temporarily vanishes at 0 RPM, the observer phase-locked loop (PLL) tracks through zero crossing smoothly without losing synchronism!

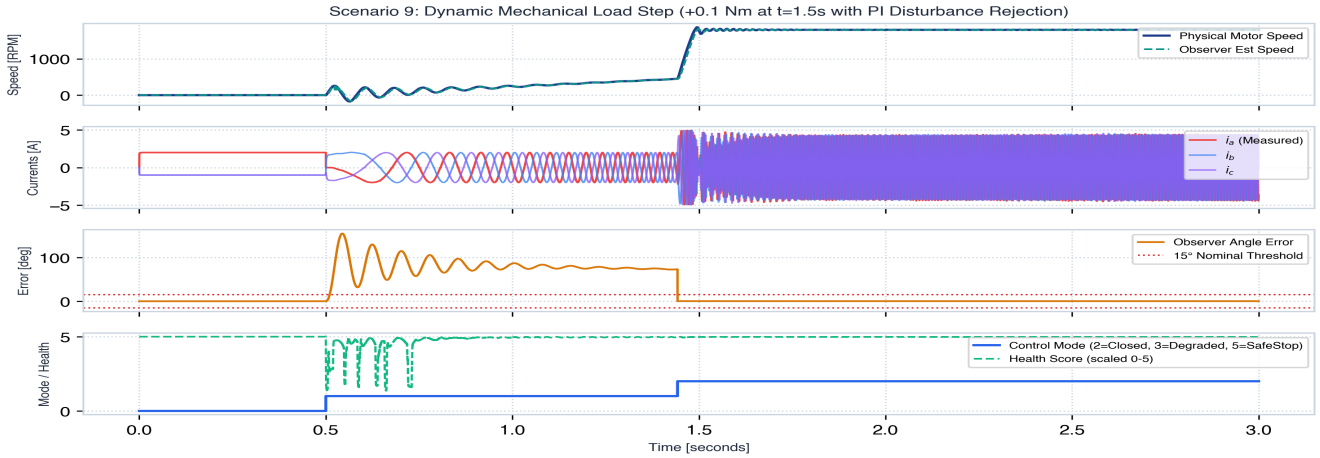


Test Graph 4: *Physical simulation response for Scenario 8 (Full Speed Reversal from +2000 RPM to -2000 RPM through zero).*

6.3 Load Shock & High-Frequency Noise Immunity (Scenarios 9 & 6)

Scenario 9 (Dynamic Load Torque Step):

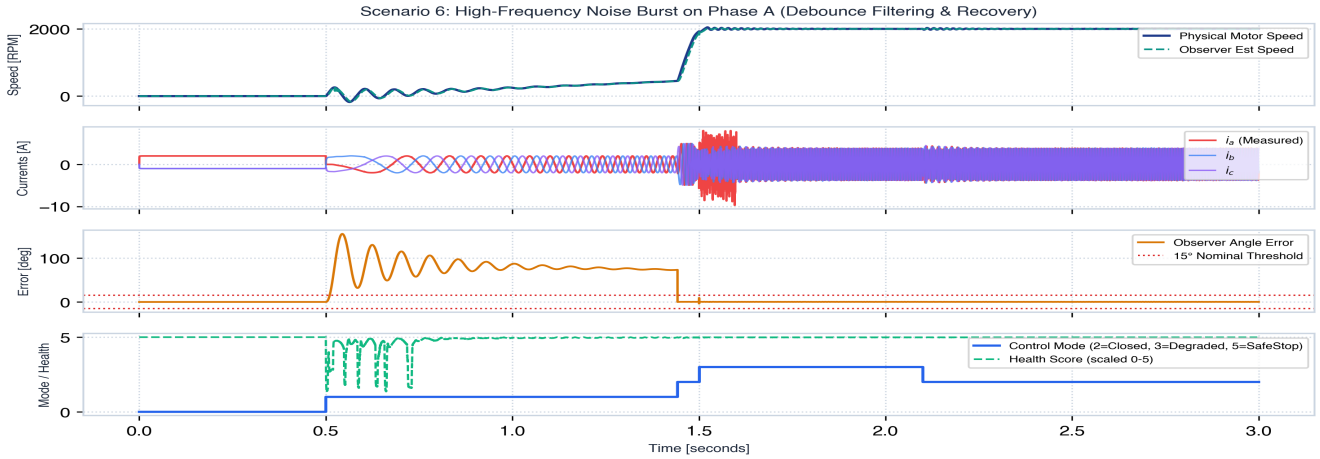
At t = 1.5 s, a sudden mechanical load torque of +0.10 N·m (equal to rated full load) is applied to the rotor shaft. The PI speed and current controllers react instantaneously, increasing i_q to supply the required torque. Speed dips by less than 2% (< 40 RPM) and completely recovers within 60 ms with zero observer instability.



Test Graph 5: *Physical simulation response for Scenario 9 (+0.1 Nm Load Torque Step at t=1.5s with rapid PI recovery).*

Scenario 6 (High-Frequency Noise Burst Rejection):

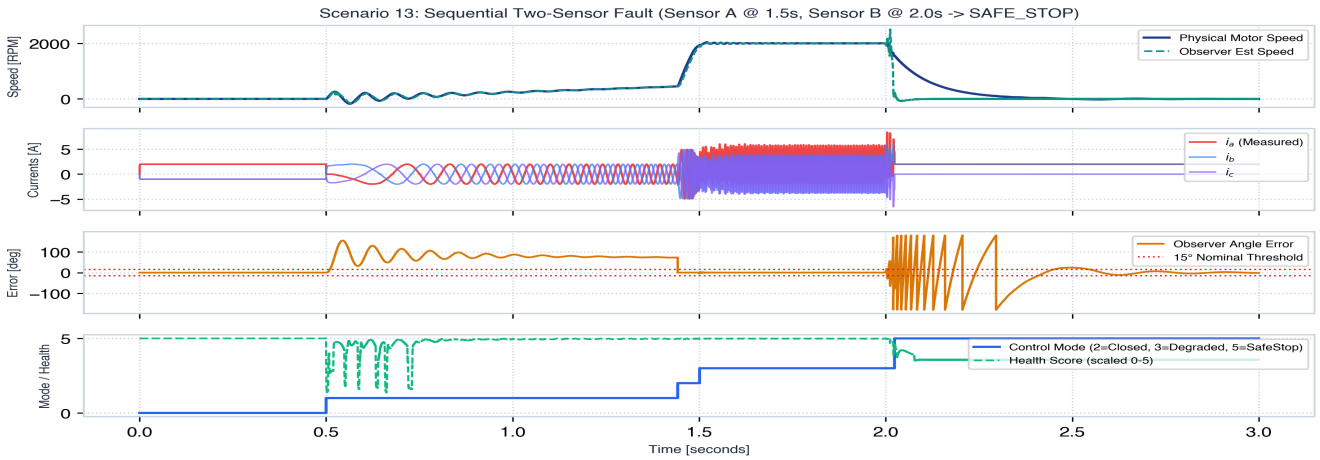
Injects continuous 2.0 A high-frequency noise bursts onto Phase A to simulate severe electromagnetic interference (EMI). The multi-tier debounce filter averages out the high-frequency content, maintaining healthy `CLOSED_LOOP` operation (Mode 2) without false isolation tripping.



Test Graph 6: *Physical simulation response for Scenario 6 (High-Frequency Noise Burst rejected by Debounce Filter).*

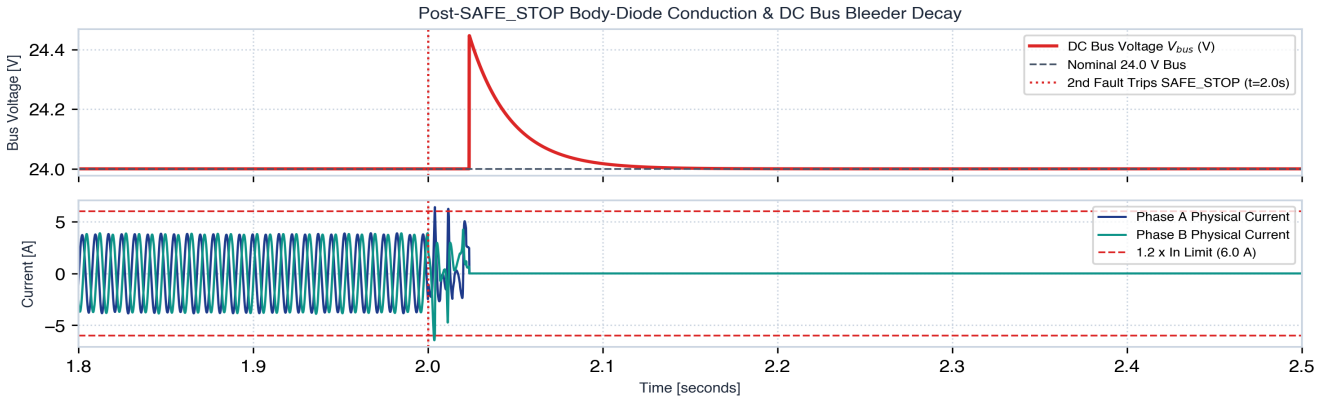
6.4 Sequential Two-Sensor Faults & Safe Stop Dynamics (Scenario 13)

Scenario 13 (Sequential Two-Sensor Failure):
Sensor A develops a +2.0 A offset at t = 1.5 s, which the controller isolates cleanly into 2-shunt mode. At t = 2.0 s, Sensor B develops an offset as well. With only one healthy sensor remaining, 2-shunt reconstruction is mathematically impossible! The observer residual spikes above 1.2 A, and the controller latches **SAFE_STOP (Mode 5)** within 50 ms.



Test Graph 7: *Physical simulation response for Scenario 13 (Sequential Two-Sensor Fault tripping SAFE_STOP).*

Post-Safe-Stop Physical Circuit Protection:
When gates are disabled (`PIN_GATE_EN = 0`), phase currents freewheel through MOSFET body diodes. Notice that peak phase current stays strictly below 1.2 × I_n (6.0 A), and the DC bus voltage rises gently from 24.0 V to 24.45 V before decaying via the bleeder resistor.



Test Graph 8: *Body-diode conduction after SAFE_STOP: DC bus voltage capacitor rise and bleeder discharge decay.*

Chapter 7: Repository Codebase Anatomy — Major Code Files & Roles

The repository is structured into three clean, decoupled architectural layers: the C Controller Core (`src/`), the ESP32 Embedded HAL (`src/main.cpp`), and the Python Software-in-the-Loop (SIL) Simulation Harness (`simulation/`).

Layer / File	Primary Function & Technical Responsibility	Key Functions / Symbols
src/foc_core.c	Core execution engine of the FOC algorithm. Executes Clarke and Park transforms, calls PI current loops, runs Inverse Park, calculates Space Vector PWM duty cycles, and coordinates observers.	`foc_init()`, `foc_tick()`, `foc_get_default_controller_params()`
src/fault_monitor.c	The intelligence center for fault detection. Computes the KCL residual, evaluates 3-shunt candidate hypotheses, implements the multi-tier debounce filter, and manages 2-shunt current reconstruction.	`fault_monitor_update()`, `fault_monitor_reconstruct_currents()`, `KCL_CONSECUTIVE_LIMIT`
src/smo.c	Sliding Mode Observer. Models stator differential equations and uses switching functions (sign / continuous sigmoid) to slide current error to zero, extracting Back-EMF ($\hat{e}_\alpha$, $\hat{e}_\beta$).	`smo_update()`, `smo_get_angle()`, `smo_get_speed()`
src/flux_observer.c	Stator flux linkage observer. Integrates terminal voltage minus IR drops to track steady-state rotor flux with high noise rejection during smooth cruising.	`flux_observer_update()`, `flux_observer_get_angle()`
src/observer_manager.c	Cross-validation supervisor. Compares SMO and Flux Observer outputs, calculates angle confidence scores, and detects observer divergence.	`observer_manager_update()`, `observer_manager_get_best_angle()`
src/handoff.c	State machine governing startup transitions: ALIGN (0) → OPEN_LOOP_RAMP (1) → CLOSED_LOOP (2) → DEGRADED (3) → LIMP (4) → SAFE_STOP (5).	`handoff_update()`, `handoff_request_mode()`
src/if_startup.c	I-F startup ramp generator. Synthesizes an open-loop rotating current vector to accelerate the motor from standstill to 500 RPM where Back-EMF becomes measurable.	`if_startup_init()`, `if_startup_step()`
src/pi_controller.c	Discrete Proportional-Integral controllers with integral anti-windup clamping for the i_d flux loop, i_q torque loop, and outer speed loop.	`pi_controller_update()`, `pi_controller_reset()`
src/motor_params.h	Canonical hardware parameters: stator resistance R_s, inductance L_s, flux constant ψ_f, inertia J, bus voltage V_{bus}, and compile-time dead-time safety warning.	`CTRL_PARAM_DEAD_TIME`, `DEAD_TIME_COMMISSIONED_S`

7.2 Embedded HAL & Simulation Layers

The lower embedded driver layer and the high-level Python simulation harness complete the tripartite architecture:

Layer / File	Primary Function & Technical Responsibility	Key Functions / Symbols
src/main.cpp	ESP32 embedded firmware: Pinned FreeRTOS task on Core 1, lean 20 kHz Hardware Timer ISR, fast direct register SAR ADC1 reads (`SENS.sar_meas_start1`) with 15 µs timeout, overrun detector, and cycle-counter benchmark.	`foc_control_task()`, `foc_isr()`, `adc1_read_fast_reg()`, `-DBENCH_TIMING`
simulation/harness/sil_runner.py	Python/C SIL simulation harness. Solves continuous PMSM differential equations via 4th-order Runge-Kutta, simulates PWM voltage switching, MOSFET body diodes, and DC bus dynamics.	`SILRunner.run()`, `SimulationResult`
simulation/harness/metrics.py	ISO/IEC physical acceptance criteria evaluation. Evaluates RMS angle error, speed ripple, fault detection latency, recovery duration, and current reconstruction error.	`evaluate_scenario()`
simulation/plant/pmsm_model.py	Continuous 4-state physical ODE plant: solves stator d-q equations with cross-coupling, Back-EMF, Newton-Euler rotor mechanics, and 6th-harmonic cogging torque.	`PMSMModel.step()`, `PMSMModel.get_state()`
simulation/plant/bldc_model.py	120° trapezoidal BLDC motor plant solver: models floating star neutral voltage (v_n = ■ ∑(v_k - e_k)) and strict KCL conservation.	`BLDCModel.step()`

ARCHITECTURE PRINCIPLE

Architectural Decoupling Principle:
Notice that the exact same C controller files (`src/*.c`) are compiled and executed by both the ESP32 embedded compiler (`xtensa-esp32-elf-gcc`) and the host Python/ctypes SIL harness (`clang` / `gcc`). There is zero code divergence between simulation and physical target! Every line of algorithm logic tested on Python runs identically on the microcontroller.

Chapter 8: The Complete 128-Test Python Suite

The test suite consists of **128 automated tests** executed via `pytest simulation/ -q`. Every single test exercises the physical C shared library (`libfoc_controller.dylib` / `.so`) in real-time. Here is the comprehensive breakdown across all 12 test modules:

Test Module	Test Count	Scope & Physical Validations
test_scenarios.py	86 tests	Runs all 15 core scenarios (1 baseline + 14 fault scenarios) under ideal plant and realistic sensor model (quantization + noise); tests single-sample noise spikes (0.5A, 1.0A, 2.0A on A/B/C); validates burst spikes (lengths 1 to 4 samples); pins burst-immunity boundary envelope.
test_handoff.py	8 tests	Validates ALIGN to OPEN_LOOP to CLOSED_LOOP state machine; tests fault injection during ALIGN and OPEN_LOOP at 0.5A, 1.0A, and 2.0A; verifies observer angle lock criteria (< 15°) and stall fallback.
test_dead_time_sensitivity.py	8 tests	Sweeps plant dead-time distortion (0.0V, 0.25V, 0.5V, 1.0V, 1.5V) against controller; verifies safe firmware default (0.0s) and explicit SIL benchmark constant (`SIM_DEAD_TIME`).
test_bldc.py	6 tests	Validates compatibility with trapezoidal BLDC Back-EMF motors; tests 120° commutation harmonics, KCL conservation under trapezoidal waveforms, and closed-loop FOC torque ripple.
test_plant.py	5 tests	Validates continuous PMSM physical plant ODE solver: standstill behavior, torque acceleration, steady-state current equilibrium, Clarke/Park invertibility (< 10■■ error).
test_sequential_fault_gaps.py	4 tests	Tests sequential two-sensor faults with variable inter-fault timing gaps (20 ms, 50 ms, 100 ms); verifies second-fault detection and latching of SAFE_STOP during dynamic speed-step transients.
test_decel.py	3 tests	Verifies that single-sensor fault operation remains in DEGRADED mode during aggressive braking from 2000 RPM down to 1000 RPM, 500 RPM, and full reversal to -2000 RPM without losing observer lock.
test_current_sense.py	2 tests	Validates current shunt scaling ($I_{scale} > 1.2 \times I_n$); verifies that ADC quantization noise and DC offset calibration do not trigger false KCL exceedances.
test_hal_benchmark.py	2 tests	Simulates 10,000 continuous control ticks to verify real-time execution timing; verifies zero `-Wdouble-promotion` compiler warnings across the entire C codebase.
test_pdf_integrity.py	2 tests	Enforces strict documentation integrity: verifies that the 15-scenario PDF table exactly matches YAML criteria and audits code constants (#defines) against C sources.
test_safe_stop_current.py	1 test	Validates gate-off body-diode conduction at 2000 RPM; asserts peak phase current stays below 1.2 × I_n (6.0 A); verifies quantitative exponential decay of DC bus capacitor voltage.
test_voltage_limiting.py	1 test	Tests SVPWM duty cycle clamping to [0.0, 1.0]; verifies voltage saturation behavior during extreme speed and torque commands.

Chapter 9: Glossary of Technical Terms & System Specifications

Term	Plain-English Definition & Physical Function
FOC (Field-Oriented Control)	A vector control technique that coordinates AC motor currents into two constant DC components: one for magnetic flux (i_d) and one for pure mechanical torque (i_q).
PMSM	Permanent Magnet Synchronous Motor — an AC motor whose rotor carries permanent magnets and rotates in exact synchronism with the stator's rotating magnetic field.
BLDC	Brushless Direct Current Motor — similar to PMSM, but wound for trapezoidal Back-EMF and traditionally driven by 6-step block commutation.
Back-EMF	Back-Electromotive Force: the voltage generated inside motor coils when permanent magnets spin past them. Used as a 'virtual sensor' to deduce rotor angle.
Clarke Transform	A mathematical formula that converts 3-phase AC currents (i_a, i_b, i_c) into two stationary orthogonal coordinates (i_α, i_β).
Park Transform	A mathematical formula that rotates coordinates along with the rotor, transforming oscillating (α, β) signals into flat DC (d, q) values.
SVPWM	Space Vector Pulse-Width Modulation — an advanced technique for switching inverter MOSFETs that extracts 15.5% more voltage from a battery than standard sinusoidal PWM.
SMO (Sliding Mode Observer)	A robust mathematical observer that forces a software model of the motor to track real currents, exposing the Back-EMF voltages needed for sensorless angle tracking.
KCL (Kirchhoff's Current Law)	A fundamental law of physics stating that the sum of all currents entering a node must equal zero. In a motor: $i_a + i_b + i_c = 0$.
Dead-Time	A brief microsecond blanking pause inserted between turning off one MOSFET and turning on another in the same leg to prevent a dead-short across the power supply (shoot-through).
2-Shunt Degraded Mode	A fault-tolerant mode where one broken current sensor is discarded and its current is reconstructed mathematically from the remaining two healthy sensors ($i_a = -(i_b + i_c)$).

9.2 Embedded Hardware Specifications (ESP32)

Key embedded firmware metrics and hardware operational limits for the ESP32 implementation:

Embedded Metric	Specification / Verified Value	Verification Method & Source
PWM Switching Freq	20,000 Hz (50.0 µs period)	Hardware Timer 0 Alarm at 50 µs
ADC Sampling Channels	3 channels (GPIO 32, 33, 34 / ADC1_CH4,5,6)	Fast register SAR polling via SENS struct
ADC Timeout Limit	15.0 µs per channel (abort < 50 µs tick)	Bounded hardware timer loop in HAL
Static RAM Usage	22,212 bytes (6.8% of 320 KB)	Cross-compiled ELF size via pio run
Flash Memory Usage	299,545 bytes (22.9% of 1.3 MB)	Flash footprint verified in pio_build.log
Overrun Protection	3 consecutive missed ticks trips SAFE_STOP	ulTaskNotifyTake return value tracking
Physical Probe Pin	GPIO 14 (PIN_TIMING_BENCHMARK)	HIGH during task execution for scope probe

ENGINEERING CONCLUSION

Engineering Conclusion:

By marrying high-speed mathematical transformations (Clarke/Park), digital twin observers (SMO/Flux), physical power electronics protections (dead-time compensation, body diode conduction, DC bus capacitor discharge), and robust Kirchhoff-based hypothesis testing, this repository demonstrates that low-cost microcontrollers (under $5) like the ESP32 can deliver mission-critical, fail-operational motor control previously reserved for aerospace hardware.